HW 1 P1 | Brief comment only

$$\vec{A} = \sum_k \sum_\alpha \frac{1}{\sqrt{2\varepsilon_0 \omega_k V}} \left(a_{k,\alpha}\, \vec{e}_{k,\alpha}\, e^{i(\vec{k}\vec{r} - \omega_k t)} + a^*_{k,\alpha}\, \vec{e}^{\,*}_{k,\alpha}\, e^{-i(kr - \omega t)} \right)$$

$$\vec{E} = -\frac{\partial A}{\partial t} \qquad \left(-i\omega_k\, a_{k\alpha}\, \vec{e}_{k\alpha}\, e^{ikx} + i\omega_k\, a^*_{k\alpha}\, \vec{e}^{\,*}_{k\alpha}\, e^{-ikx} \right)$$

$$\vec{B} = \vec{\nabla} \times \vec{A} = \qquad \left(i\vec{k} \times \vec{e}_{k\alpha}\, a_{k\alpha}\, e^{ikx} - i\vec{k} \times \vec{e}^{\,*}_{k\alpha}\, a^*_{k\alpha}\, e^{-ikx} \right)$$

$$\vec{E} \times \vec{B} \sim \int d\vec{r} \sum_{kk'} \sum_{\alpha\alpha'} \frac{1}{2\varepsilon_0 V \sqrt{\omega_k \omega_{k'}}} \Big(\omega_k\, a_{k\alpha}\, \vec{e}_{k\alpha} \times \vec{k} \times \vec{e}_{k'\alpha'}\, a_{k'\alpha'}$$

$$+ \omega_k\, a^*_{k\alpha}\, \vec{e}^{\,*}_{k\alpha} \times \vec{k}' \times \vec{e}^{\,*}_{k'\alpha'}\, a^*_{k'\alpha'}\, e^{-i(k+k')r + i(\omega_k + \omega_{k'})t}$$

$$- \omega_k\, a_{k\alpha}\, \vec{e}_{k\alpha} \times \vec{k}' \times \vec{e}^{\,*}_{k'\alpha'}\, a^*_{k'\alpha'}\, e^{i(k-k')r - i(\omega_k - \omega_{k'})t}$$

$$- \omega_k\, a^*_{k\alpha}\, \vec{e}^{\,*}_{k\alpha} \times \vec{k}' \times \vec{e}_{k'\alpha'}\, a_{k\alpha}\, e^{-i(k-k')r + i(\omega_k - \omega_{k'})t} \Big)$$

① $$\sum_k \sum_{\alpha\alpha'} \frac{\omega_k}{\sqrt{\omega^2_k}}\, a^*_{k\alpha}\, a^*_{-k\alpha'}\, \vec{e}^{\,*}_{k\alpha} \times (-\vec{k}) \times \vec{e}^{\,*}_{-k\alpha'}\, e^{i2\omega_k t}$$

$$a \times (b \times c) = b(a \cdot c) - c(a \cdot b) \Rightarrow -\vec{k} \underbrace{\left(\vec{e}^{\,*}_{k\alpha} \cdot \vec{e}^{\,*}_{-k\alpha'} \right)}_{\delta_{\alpha\alpha'}} + \vec{e}^{\,*}_{-k\alpha'} \left(\vec{e}^{\,*}_{k\alpha} \cdot \vec{k} \right)$$

$$\vec{k} \cdot \vec{e}_{k,\alpha} = 0$$

$$= \sum_{\vec{k}} \sum_\alpha \vec{k}\, \cancel{\omega_k}\, a^*_{k\alpha}\, a^*_{-k\alpha}\, e^{i2\omega_k t}$$

$$(\vec{k} = -\vec{p}) \quad = \sum_\alpha \sum -\vec{p}\, \cancel{\omega_p}\, a^*_{-p\alpha}\, a^*_{p\alpha}\, e^{i2\omega_p t} \quad \Rightarrow 0$$

② While 'normal' terms give $\sum_{\alpha\vec{k}} \vec{k}\, \cancel{\omega_k}\, a^*_{k\alpha}\, a_{k\alpha}$

and $\vec{k} \to -\vec{k}$ does not nullify it because $a^*_{k\alpha}\, a_{k\alpha} \neq a^*_{-k\alpha}\, a_{-k\alpha}$

it is not a symmetry operation!

This is crucial difference between "anomalous" terms (like $a^*_k a^*_{-k}$) and "normal" ones (like $a^*_k a_k$)
→ and a common mistake!